

Week 5 Report: Data Preprocessing (end) and Model Development (start)

Project: Analysing the Role of Big Data Analytics for Cyber Threat Detection and Prevention in Banking Organisations of Ireland

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What I did this week

This week I finished preprocessing and built the first models. As planned in my proposal I use Decision Tree, Random Forest and SVM. I also added Logistic Regression and Gradient Boosting as reference models, and a simple rule so I can see how much machine learning adds.

The rule is learnt from the training data only. It flags a transaction if its merchant has any suspicious history in training and the amount is above 197,827. 17 merchants count as risky. On the test set this rule already reaches an F1 of 0.995, which is very close to the machine learning models. This is an important result for my research question on effectiveness.

Next week

Tune each model with cross-validation on the training split.